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# Answer-Conditioned Chain-of-Thought Distillation for Few-Shot Industrial Vision with Small VLMs

BMVC 2026 Submission # 995

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## Abstract

Deploying AI-based visual inspection in manufacturing is challenging because production requirements change frequently, new defect types emerge, and large labeled datasets are rarely available on the factory floor. Traditional vision models require thousands of annotated images and lengthy retraining cycles, making them impractical when only a handful of expert-labeled samples exist. We propose answer-conditioned chain-of-thought (CoT) distillation as a method for rapidly adapting small vision-language models (VLMs) to new industrial tasks using minimal labeled data. A frontier VLM is given each training image along with its correct label and generates a justified visual explanation describing why the classification is correct. A 3B-parameter model is then fine-tuned on these reasoning-augmented examples via LoRA. By conditioning on correct answers, we ensure that all training reasoning is directed toward the correct conclusion, which is critical because the frontier model’s own classification accuracy ranges from 29.6% to 91.1% across our tasks. We validate the method on four industrial classification tasks spanning three image modalities (macro photography, optical microscopy, and X-ray radiography) using only 18 to 30 labeled images per task. Our method consistently outperforms direct fine-tuning by 0.9 to 8.3 percentage points across all four domains. On weld radiograph defect classification, the fine-tuned 3B model outperforms the frontier model by 10.8 percentage points using just 24 training images. On concrete aggregate grading, a task with 9 classes, the model achieves 79.6% accuracy from only 18 images where the base model scores 12.0% and the frontier model scores 29.6%. These results demonstrate that answer-conditioned CoT distillation enables practical industrial deployment of VLMs in data-scarce environments where collecting large datasets is infeasible due to cost, rarity of defects, or rapidly changing inspection criteria.

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## 1 Introduction

Visual quality inspection is central to manufacturing. Every production line needs to identify defects, classify materials, or verify compliance with standards. In practice, these inspection requirements change often. A factory may switch products, encounter new defect types, or adopt updated grading standards. Each change demands a new or retrained vision model. Traditional approaches rely on convolutional neural networks trained on thousands of labeled images [2]. Collecting this data is expensive. Expert annotators are scarce. Some

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